	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering**	
	Assignment No : 3	

Study Jenkins Architecture and Plugins with Jenkins Installation and Setup.

Objective: To study the architecture of Jenkins, understand the role and usage of Jenkins plugins, and perform the installation and basic setup of Jenkins in order to gain practical knowledge of Continuous Integration and Continuous Deployment (CI/CD) concepts.

Software Requirements:

Operating System: Windows / Linux / macOS
Java JDK (Version 17,21 or above)
Jenkins (Latest Stable Version)
Web Browser (Chrome / Firefox)

Theory

1. CI/CD Overview

Continuous Integration (CI) is the process of frequently integrating code into a shared repository with automated testing. Continuous Deployment (CD) automatically deploys code to production after successful testing. CI/CD helps in faster, reliable, and automated software delivery.

2. Jenkins Overview

Jenkins is an open-source automation tool used for CI/CD. It automates building, testing, and deployment of applications and supports multiple plugins for integration.

3. Jenkins Architecture

Jenkins uses a master-agent architecture. The master manages jobs and distributes tasks, while agents execute the assigned tasks. This improves performance and scalability.

4. Jenkins Plugins

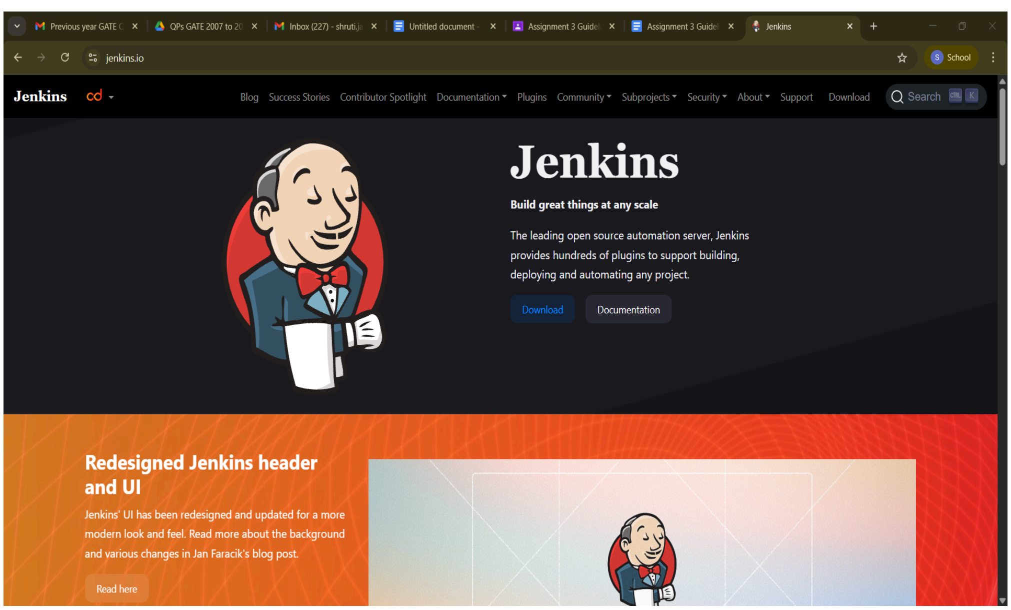
Plugins extend Jenkins functionality. They help integrate tools like Git, Docker, and Maven, making Jenkins flexible and customizable.

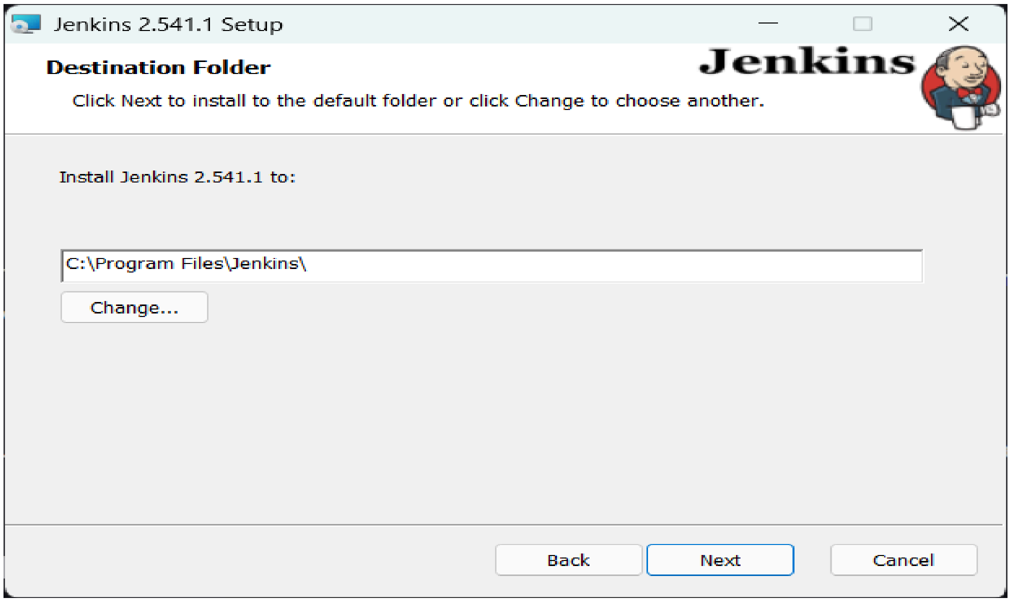
5. Jenkins Installation Procedure

Install Java, download Jenkins, run the installer, and start Jenkins on a browser. Unlock Jenkins, install plugins, create an admin user, and begin using it.

Procedure for Jenkins Installation :
(Note: Add screenshot after every stage)

1. Install Java JDK on the system and set up environmental variable.
2. Download Jenkins from the official Jenkins website.
3. Install Jenkins and start the Jenkins service.

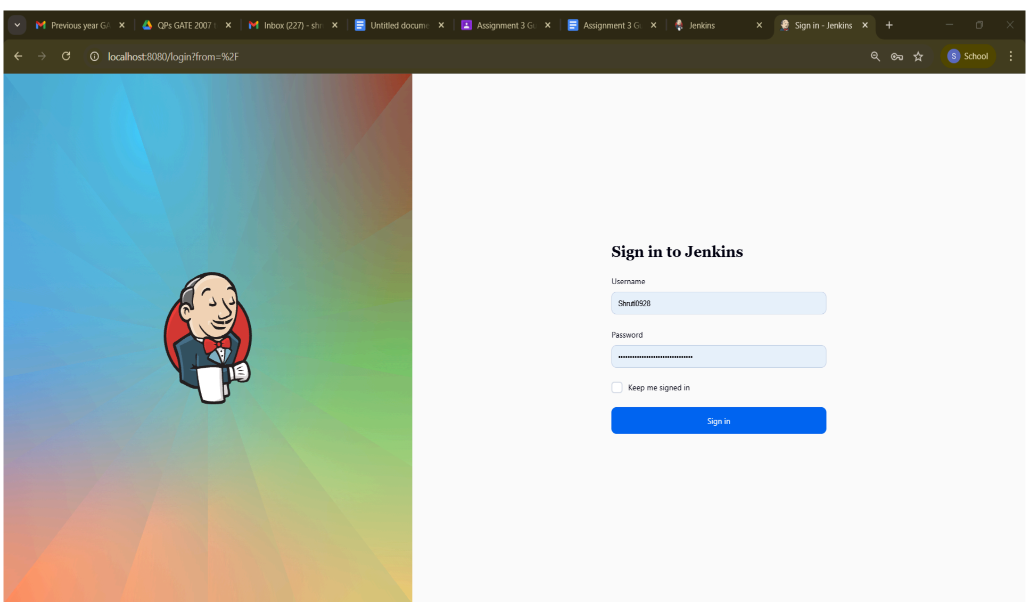


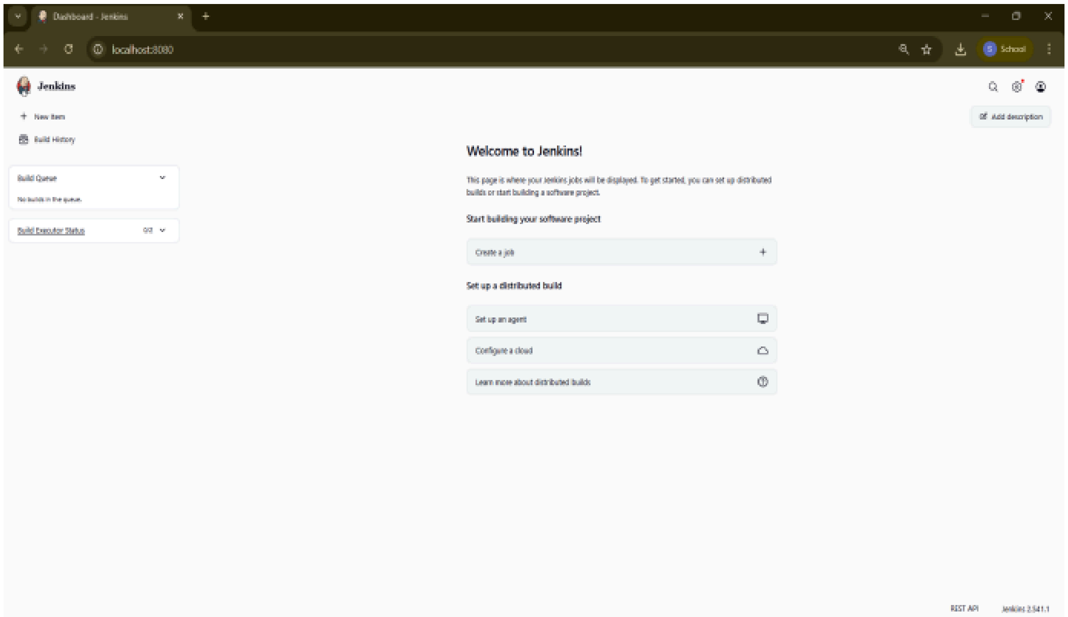


4. Install Jenkins and start the Jenkins service

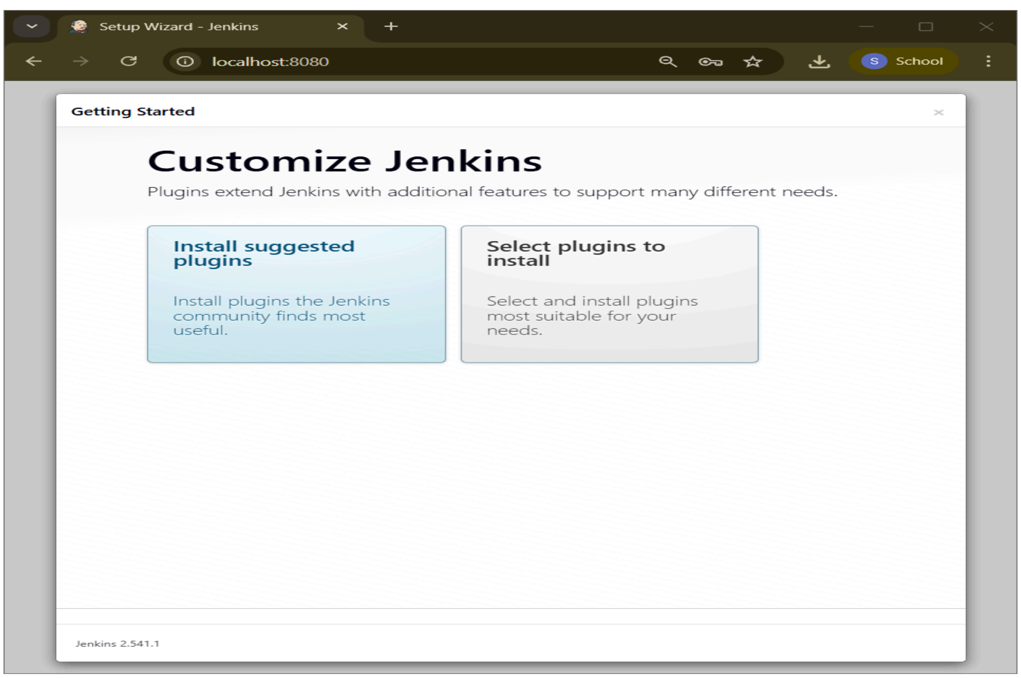
5. Access Jenkins through a web browser using http://localhost:8080.

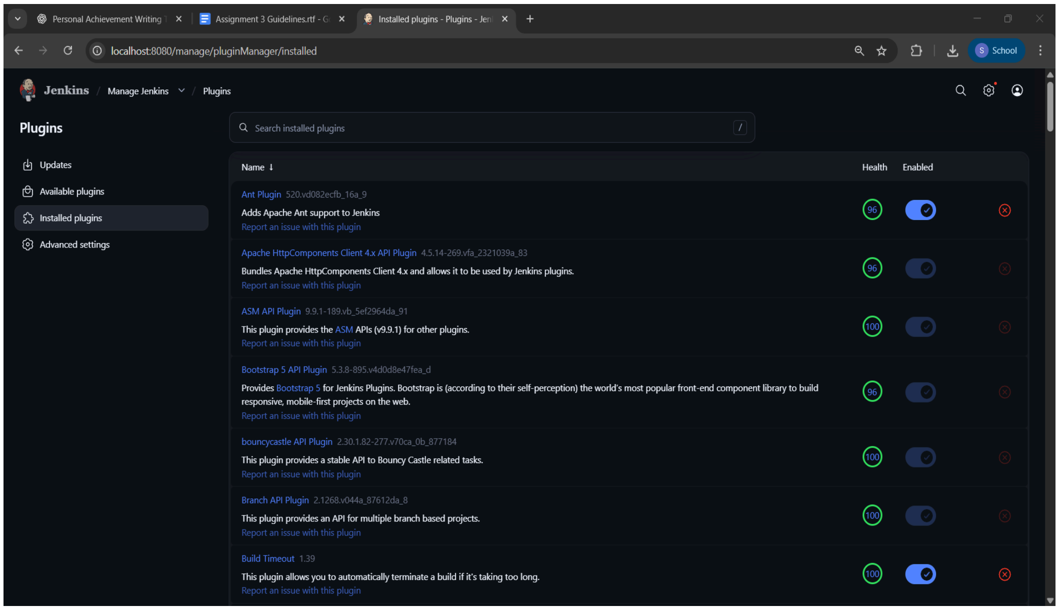
6. Unlock Jenkins using the administrator password.



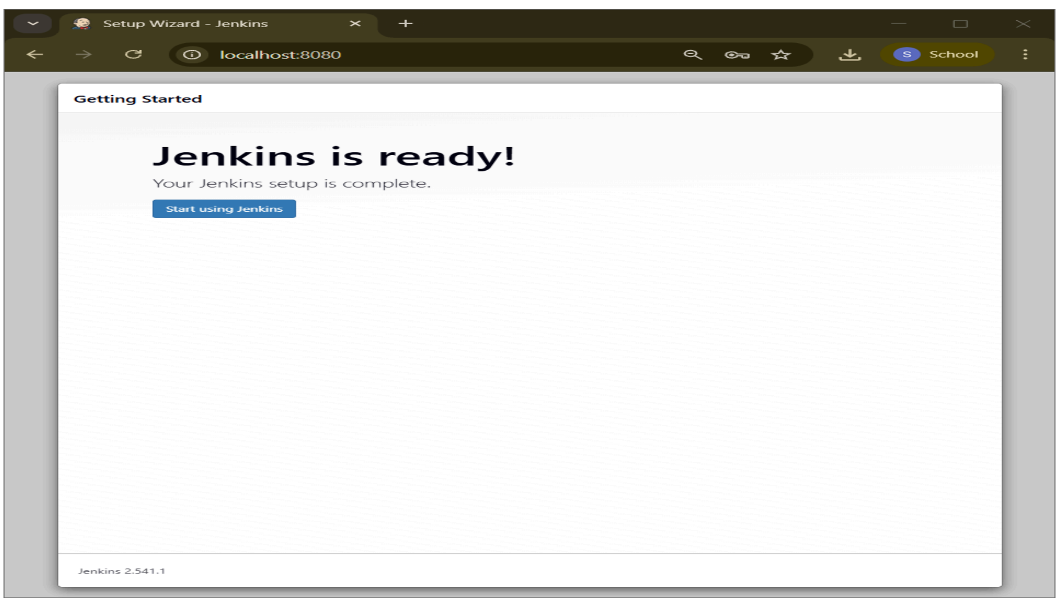


7. Install suggested plugins.

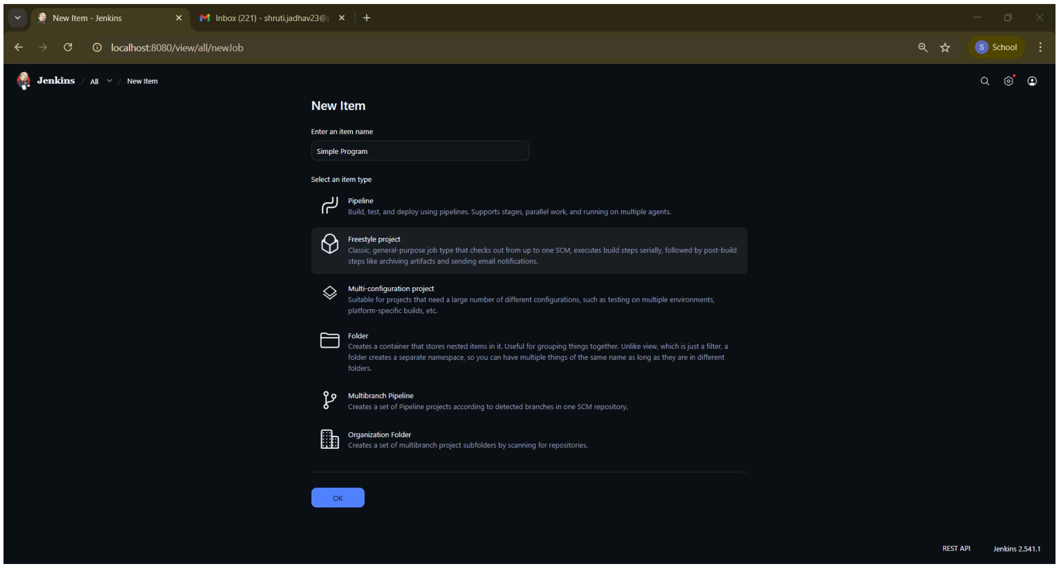


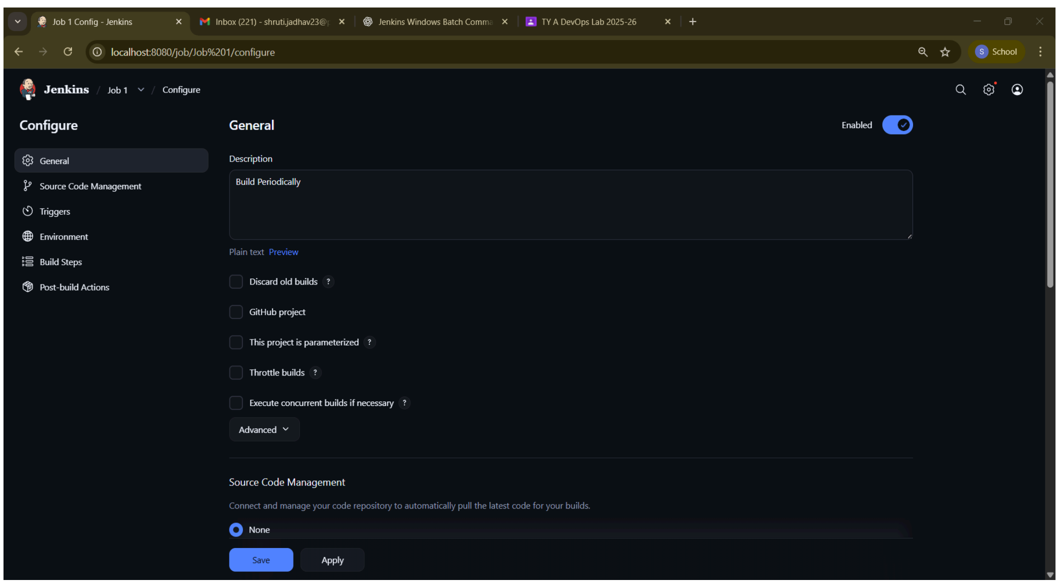


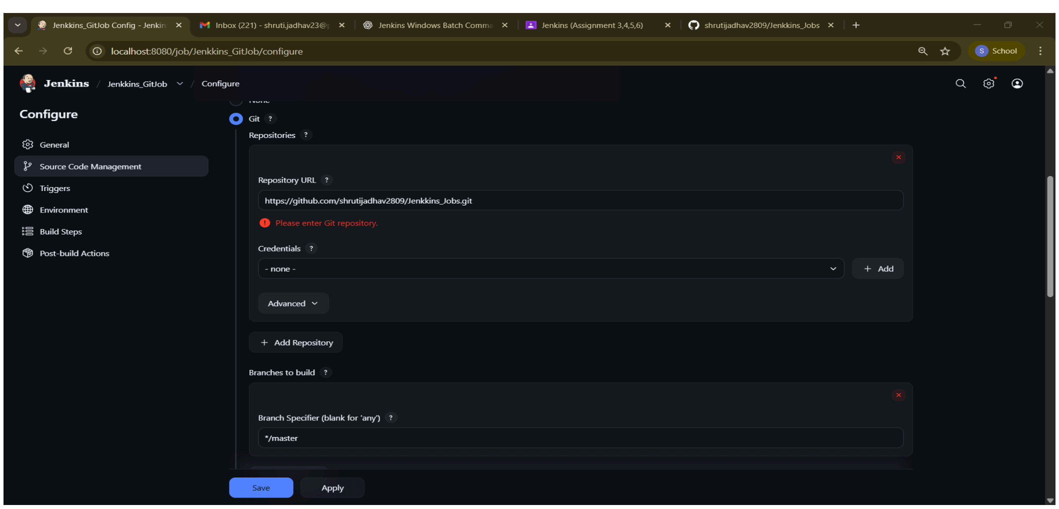
8. Create an admin user and complete setup.

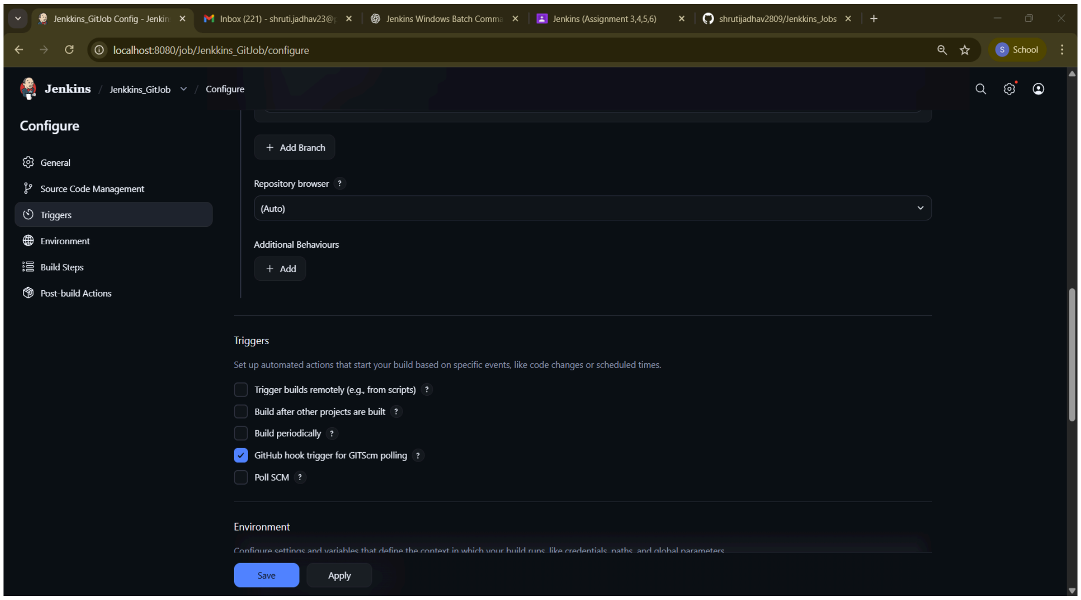


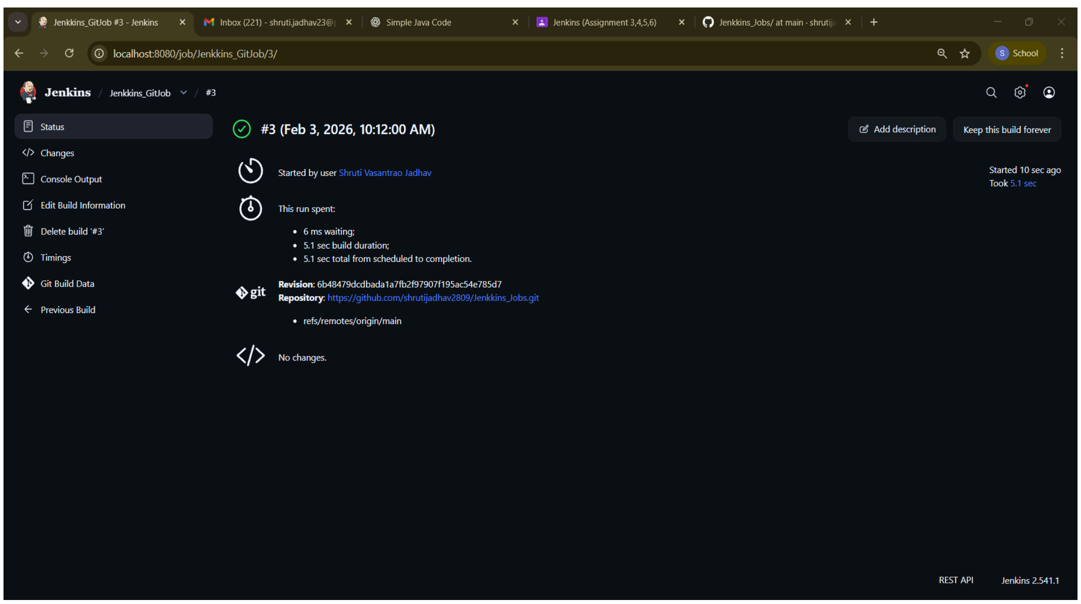
9. Create a new Jenkins job and execute a sample build.

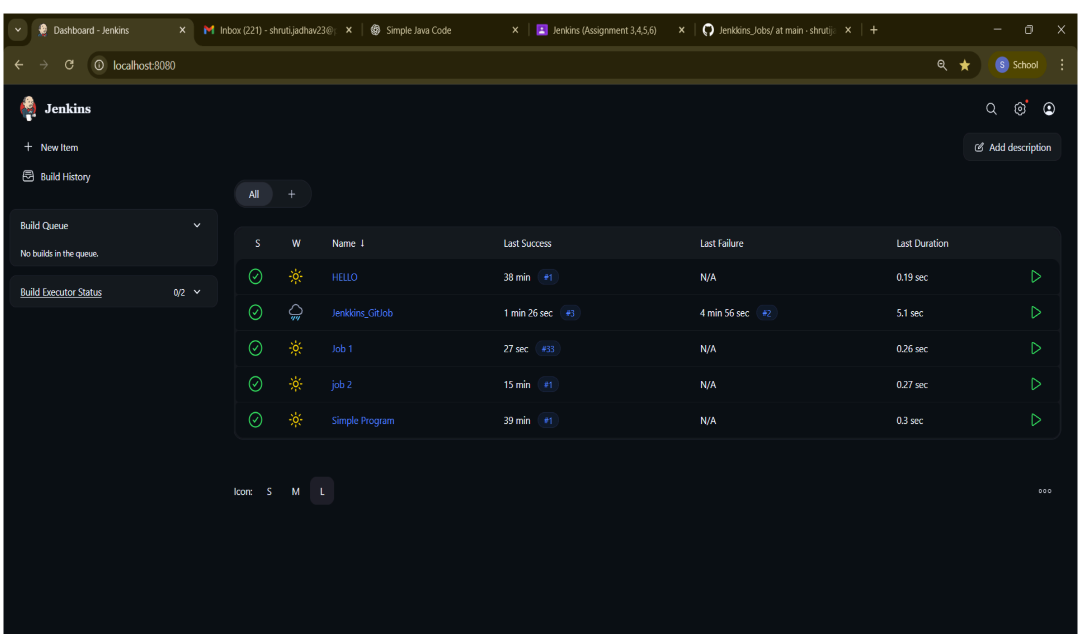

New Item - Jenkins
localhost:8080/view/all/newJob
Jenkins
New Item
New Item
Enter an item name
Simple Program
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
OK
REST API
Jenkins 2.541.1


Job 1 Config - Jenkins
localhost:8080/job/Job%201/configure
Jenkins
Job 1
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
General
Enabled
Description
Build Periodically
Plain text Preview
Discard old builds
GitHub project
This project is parameterized
Throttle builds
Execute concurrent builds if necessary
Advanced
Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.
None
Save
Apply


Jenkkins_GitJob Config - Jenkins
localhost:8080/job/Jenkkins_GitJob/configure
Jenkins
Jenkkins_GitJob
Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Git
Repositories
Repository URL
https://github.com/shrutijadhav2809/Jenkkins_Jobs.git
Please enter Git repository.
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Branch Specifier (blank for 'any')
*/master
Save
Apply


localhost:8080/job/Jenkkins_GitJob/configure
Jenkins / Jenkkins_GitJob / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Add Branch
Repository browser
(Auto)
Additional Behaviours
Add
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Trigger builds remotely (e.g., from scripts)
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Environment
Save
Apply


localhost:8080/job/Jenkkins_GitJob/3/
Jenkins / Jenkkins_GitJob / #3
Status
Changes
Console Output
Edit Build Information
Delete build '#3'
Timings
Git Build Data
Previous Build
#3 (Feb 3, 2026, 10:12:00 AM)
Add description
Keep this build forever
Started by user Shruti Vasantrao Jadhav
Started 10 sec ago
Took 5.1 sec
This run spent:
6 ms waiting;
5.1 sec build duration;
5.1 sec total from scheduled to completion.
Revision: 6b48479dcdbada1a7fb2f97907f195ac54e785d7
Repository: https://github.com/shrutijadhav2809/Jenkkins_Jobs.git
refs/remotes/origin/main
No changes.
REST API
Jenkins 2.541.1


localhost:8080
Jenkins
New Item
Build History
Add description
Build Queue
No builds in the queue.
Build Executor Status 0/2
All
S W Name ↓ Last Success Last Failure Last Duration
HELLO 38 min #1 N/A 0.19 sec
Jenkkins_GitJob 1 min 26 sec #3 4 min 56 sec #2 5.1 sec
Job 1 27 sec #33 N/A 0.26 sec
job 2 15 min #1 N/A 0.27 sec
Simple Program 39 min #1 N/A 0.3 sec
Icon: S M L

FAQs:

1. What is the difference between Continuous Delivery and Continuous Deployment?

Aspect	Continuous Delivery	Continuous Deployment
Automation Level	Partially automated with a manual approval step	Fully automated with no manual intervention
Deployment Frequency	Typically, less frequent, scheduled releases	Very frequent, potentially multiple times a day
Risk Management	Allows for manual checks before deployment	Relies heavily on automated testing
Use Case	Suitable for environments requiring compliance or high-stakes releases	Ideal for environments needing rapid iteration and user feedback

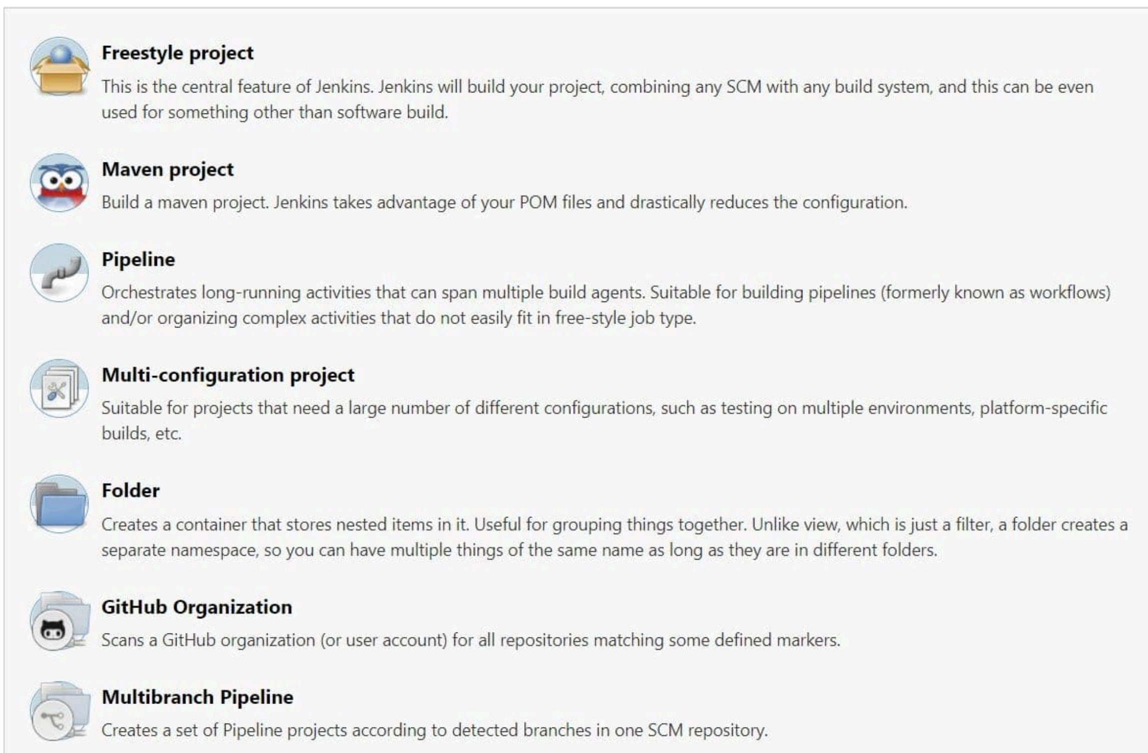
2. What are the different types of jobs available in Jenkins?

Jenkins provides the following main types of jobs:

1. **Freestyle Project**
    - Basic and most commonly used job
    - Easy to configure using UI
2. **Pipeline Job**
    - Uses a Jenkinsfile
    - Supports CI/CD pipelines
3. **Multibranch Pipeline**

- Automatically builds multiple branches
- Used with Git repositories

4. **Maven Project**
    - Specifically for Maven-based Java projects
5. **External Job**
    - Used to monitor external processes



3. Explain Freestyle job in jenkins.

A Freestyle Job is the simplest and most basic job in Jenkins.

Key Points:

- It is created using Jenkins web interface
- No coding is required to configure it
- Supports many build steps like:
  - Execute Windows batch command
  - Execute shell
  - Run Maven, Ant, or Gradle
- Can be triggered manually or automatically
- Supports Git integration and plugins

Example:

A freestyle job can:

- Pull code from GitHub
- Compile the code
- Run a script
- Generate output

Conclusion:

Jenkins supports DevOps practices by automating software build and deployment processes. Continuous Delivery ensures the application is always ready for release with manual approval, while Continuous Deployment automates the release completely. Jenkins provides various job types, with Freestyle jobs being the simplest and easiest to use for basic automation tasks.